



Computer Systems Administration (IE2060)
2nd Year, 2nd Semester

Assignment

Caching Proxy with SSL

Submitted to
Sri Lanka Institute of Information Technology

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In partial fulfillment of the requirements for the
Bachelor of Science Special Honors Degree in Information Technology

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1) Installed and started squid (During Apache Lab I installed open ssl).

```
[root@mlb-dcl-centos9 ~]# dnf install -y squid
Extra Packages for Enterprise Linux 9 - x86_64 919 kB/s | 23 MB 00:26
Extra Packages for Enterprise Linux 9 openh264 233 B/s | 2.5 kB 00:10
Extra Packages for Enterprise Linux 9 - Next - 50 kB/s | 615 kB 00:12
Dependencies resolved.
=====
Package Architecture Version Repository Size
=====
Installing:
squid x86_64 7:5.5-20.el9 appstream 3.7 M
Installing dependencies:
libecap x86_64 1.0.1-10.el9 appstream 26 k
perl-Digest-SHA x86_64 1:6.02-461.el9 appstream 62 k
perl-English noarch 1.11-481.el9 appstream 13 k
Transaction Summary
=====
Install 4 Packages

Total download size: 3.8 M
Installed size: 11 M
Downloading Packages:
```

```
[root@mlb-dcl-centos9 ~]# squid -v
Squid Cache: Version 5.5
Service Name: squid

This binary uses OpenSSL 3.5.0 8 Apr 2025. For legal restrictions on distribution see https://www.openssl.org/source/license.html

configure options: '--build=x86_64-redhat-linux-gnu' '--host=x86_64-redhat-linux-gnu' '--program-prefix=' '--prefix=/usr' '--exec-prefix=/usr' '--bindir=/usr/bin' '--sbindir=/usr/sbin' '--sysconfdir=/etc' '--datadir=/usr/share' '--includedir=/usr/include' '--libdir=/usr/lib64' '--libexecdir=/usr/libexec' '--localstatedir=/var' '--sharedstatedir=/var/lib' '--mandir=/usr/share/man' '--infodir=/usr/share/info' '--libexecdir=/usr/lib64/squid' '--datadir=/usr/share/squid' '--sysconfdir=/etc/squid' '--with-logdir=/var/log/squid' '--with-pidfile=/run/squid.pid' '--disable-dependency-tracking' '--enable-eui' '--enable-follow-x-forwarded-for' '--enable-auth' '--enable-auth-basic=DB,fake,getpwnam,LDAP,NCSA,PAM,POP3,RADIUS,SASL,SMB,SMB_LM' '--enable-auth-ntlm=SMB_LM,fake' '--enable-auth-digest=file,LDAP' '--enable-auth-negotiate=kerberos' '--enable-external-acl-helpers=LDAP_group,time_quota,session,unix_group,wbinfo_group,kerberos_ldap_group' '--enable-storeid-rewrite-helpers=file' '--enable-cache-digests' '--enable-cachemgr-hostname=localhost' '--enable-delay-pools' '--enable-epoll' '--enable-icap-client' '--enable-ident-lookups' '--enable-linux-netfilter' '--enable-removal-policies=heap,lru' '--enable-snpmp' '--enable-ssl' '--enable-ssl-crt' '--enable-storeio=aufs,diskd,ufs,rock' '--enable-diskio' '--enable-wccpv2' '--disable-esi' '--enable-ecap'
```

```
[root@mlb-dcl-centos9 ~]# systemctl status squid
● squid.service - Squid caching proxy
   Loaded: loaded (/usr/lib/systemd/system/squid.service; enabled; preset: disabled)
   Active: active (running) since Thu 2025-05-08 11:19:05 +0530; 15s ago
```

2) Set firewall permission for port 3128 in server

```
[root@mlb-dcl-centos9 ~]# firewall-cmd --permanent --add-port=3128/tcp
success
[root@mlb-dcl-centos9 ~]# firewall-cmd --reload
success
```

3) Doing basic configurations of squid (my network – 192.168.1.0/24).

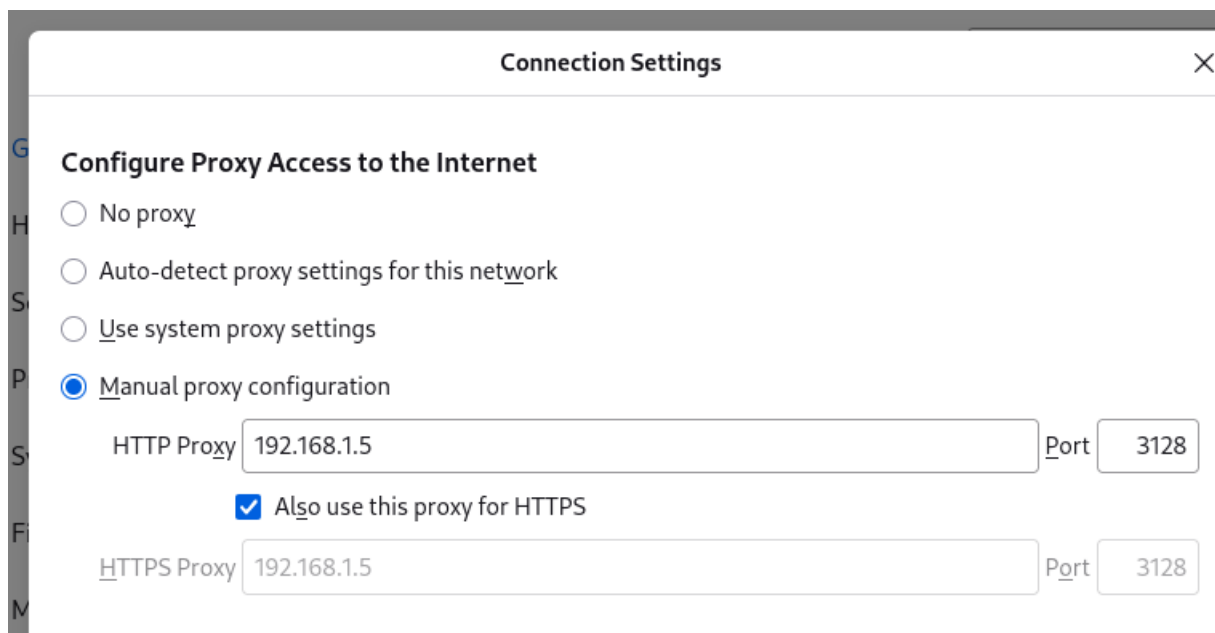
```
# Example rule allowing access from your local networks.
# Adapt to list your (internal) IP networks from where browsing
# should be allowed
acl localnet src 192.168.1.0/24
acl localhost src 127.0.0.1/32 ::1
```

```
# from where browsing should be allowed
http_access allow localnet
http_access allow localhost

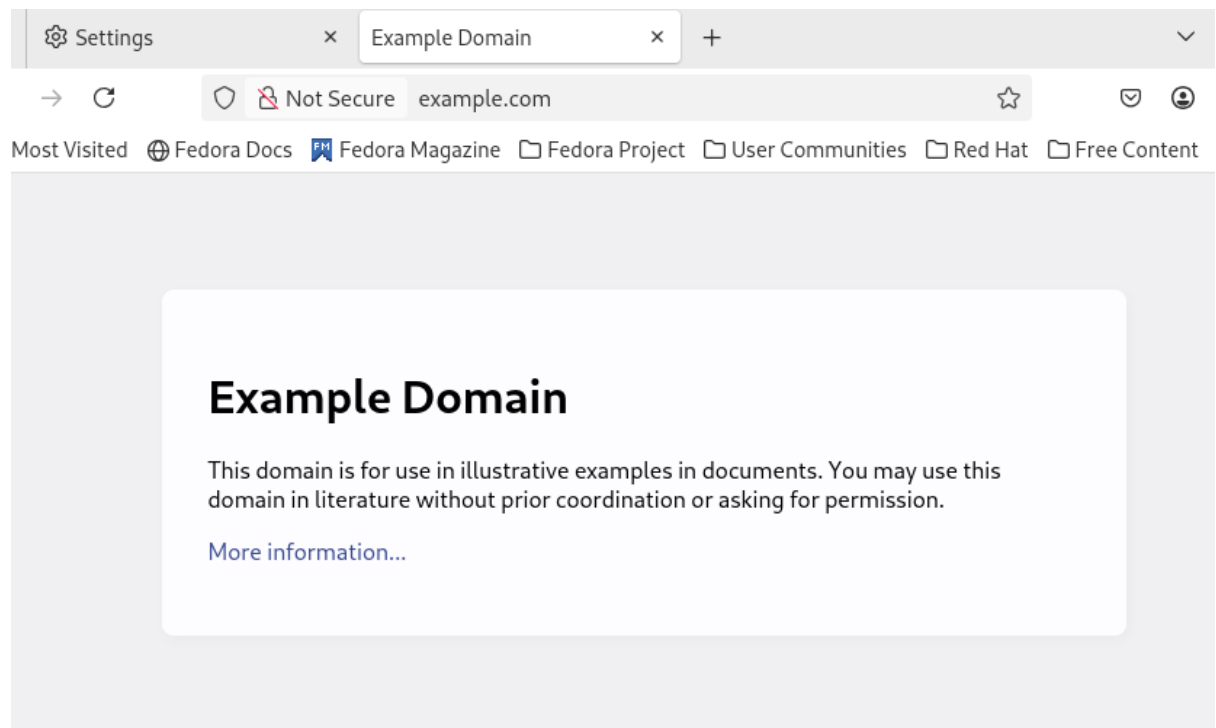
# And finally deny all other access to the proxy
http_access deny all

# Squid normally listens to port 3128
http_port 3128
```

4) Setting port in fedora firefox browser settings



5) Now able to go http sites from browser



6) Also can curl from terminal

```
root@fedora:~ — -bash
root@fedora:~# export http_proxy="http://192.168.1.5:3128"
root@fedora:~# curl http://example.com
<!doctype html>
<html>
<head>
  <title>Example Domain</title>

  <meta charset="utf-8" />
  <meta http-equiv="Content-type" content="text/html; charset=utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <style type="text/css">
  body {
    background-color: #f0f0f2;
    margin: 0;
    padding: 0;
    font-family: -apple-system, system-ui, BlinkMacSystemFont, "Segoe UI", "
Open Sans", "Helvetica Neue", Helvetica, Arial, sans-serif;
  }
  div {
    width: 600px;
    margin: 5em auto;
    padding: 2em;
    background-color: #fdfdff;
```

- 7) Setting caching directory and replacement policy (This is temporary and I am changing these settings later. (LRU -> LFUDA)

```
# Squid normally listens to port 3128
http_port 3128

# Uncomment and adjust the following to add a disk cache directory.
cache_dir ufs /var/spool/squid 100 16 256
cache_replacement_policy lru
# Leave coredumps in the first cache dir
coredump_dir /var/spool/squid
```

- 8) Initializing the caching directory from the command “squid -z”

```
o keep splay tree searching predictable
2025/05/08 11:44:15 kid1| WARNING: You should probably remove '192.168.1.0/24' from the ACL named 'localnet'
2025/05/08 11:44:15 kid1| Set Current Directory to /var/spool/squid
2025/05/08 11:44:15 kid1| Creating missing swap directories
2025/05/08 11:44:15 kid1| /var/spool/squid exists
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/00
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/01
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/02
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/03
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/04
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/05
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/06
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/07
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/08
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/09
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/0A
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/0B
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/0C
2025/05/08 11:44:15 kid1| Making directories in /var/spool/squid/0D
2025/05/08 11:44:16 kid1| Making directories in /var/spool/squid/0E
2025/05/08 11:44:16 kid1| Making directories in /var/spool/squid/0F
2025/05/08 11:44:16| Removing PID file (/run/squid.pid)
```

- 9) caching behavior of http sites

```
text/html
1746686452.559      528 192.168.1.6 TCP_MISS/200 528 GET http://lushbrightsilversu
nrise.neverssl.com/favicon.ico - HIER_DIRECT/2600:1f13:37c:1400:ba21:7165:5fc7:7
36e image/vnd.microsoft.icon
1746686458.906      0 192.168.1.6 TCP_INM_HIT/304 325 GET http://lushbrightsilve
rsunrise.neverssl.com/online/ - HIER_NONE/- text/html
1746686466.106      0 192.168.1.6 TCP_INM_HIT/304 326 GET http://lushbrightsilve
rsunrise.neverssl.com/online/ - HIER_NONE/- text/html
1746686471.210      0 192.168.1.6 TCP_INM_HIT/304 326 GET http://lushbrightsilve
rsunrise.neverssl.com/online/ - HIER_NONE/- text/html
```

```
1746643712.066      1 192.168.1.100 TCP_HIT/200 1129 GET http://example.co
m/ - HIER_NONE/- text/html
```

```
1745825491.463     1192 192.168.1.100 TCP_MISS/200 102785 GET http://speedtest.tel
e2.net/100KB.zip - HIER_DIRECT/90.130.70.73 application/zip
1745825516.149      1 192.168.1.100 TCP_MEM_HIT/200 102792 GET http://speedtest.
tele2.net/100KB.zip - HIER_NONE/- application/zip
```

10) Adding DNS record for proxy

```

$TTL 86400
@ IN SOA mlb-dc1-centos9.csa.lk. root.csa.lk. (
    2025050801 ;Serial
    3600        ;Refresh
    1800        ;Retry
    604800      ;Expire
    86400       ;Minimum TTL
)

@      IN NS      mlb-dc1-centos9.csa.lk.
mlb-dc1-centos9 IN A      192.168.1.5
@      IN A       192.168.1.5
@      IN A       192.168.1.100
fedora IN A       192.168.1.100
www    IN A       192.168.1.5
proxy  IN A       192.168.1.5
~
~
~
~
~
~
~
~
~
~
<named/forward.csa.lk" 16L, 380B      1,1      All

```

11) Testing from fedora client.

```
root@fedora:~# nslookup proxy.csa.lk
Server:      192.168.1.5
Address:     192.168.1.5#53

Name:   proxy.csa.lk
Address: 192.168.1.5

root@fedora:~# dig proxy.csa.lk

; <<> DiG 9.18.33 <<> proxy.csa.lk
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 11533
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; COOKIE: 0135800166e7632a01000000681c675695c2d9a917f8d63a (good)
;; QUESTION SECTION:
;proxy.csa.lk.                IN      A
```

12) Implementing SSL Bumping in squid. Creating the Cert generation directory & setting ownership.

```
[root@mlb-dc1-centos9 ~]# sudo /usr/lib64/squid/security_file_certgen -c -s /var/lib/ssl_db -M 4MB
Initialization SSL db...
Done
[root@mlb-dc1-centos9 ~]# sudo chown -R squid:squid /var/lib/ssl_db
[root@mlb-dc1-centos9 ~]#
```

13) Editing squid.conf file to enable SSL Bump

```
root@mlb-dc1-centos9:~
#
# Recommended minimum configuration:
#
# Example rule allowing access from your local networks.
# Adapt to list your (internal) IP networks from where browsing
# should be allowed
acl localnet src 192.168.1.0/24
acl localhost src 127.0.0.1/32 ::1
acl localnet src 0.0.0.1-0.255.255.255 # RFC 1122 "this" network (LAN)
acl localnet src 10.0.0.0/8 # RFC 1918 local private network (LAN)
acl localnet src 100.64.0.0/10 # RFC 6598 shared address space (CGN)
acl localnet src 169.254.0.0/16 # RFC 3927 link-local (directly plugged) machines
acl localnet src 172.16.0.0/12 # RFC 1918 local private network (LAN)
acl localnet src 192.168.0.0/16 # RFC 1918 local private network (LAN)
acl localnet src fc00::/7 # RFC 4193 local private network range
acl localnet src fe80::/10 # RFC 4291 link-local (directly plugged) machines

acl SSL_ports port 443
acl Safe_ports port 80 # http
acl Safe_ports port 21 # ftp
acl Safe_ports port 443 # https
acl Safe_ports port 70 # gopher
```

```
root@mlb-dc1-centos9:~
acl localnet src 169.254.0.0/16 # RFC 3927 link-local (directly plugged) machines
acl localnet src 172.16.0.0/12 # RFC 1918 local private network (LAN)
acl localnet src 192.168.0.0/16 # RFC 1918 local private network (LAN)
acl localnet src fc00::/7 # RFC 4193 local private network range
acl localnet src fe80::/10 # RFC 4291 link-local (directly plugged) machines

acl SSL_ports port 443
acl Safe_ports port 80 # http
acl Safe_ports port 21 # ftp
acl Safe_ports port 443 # https
acl Safe_ports port 70 # gopher
acl Safe_ports port 210 # wais
acl Safe_ports port 1025-65535 # unregistered ports
acl Safe_ports port 280 # http-mgmt
acl Safe_ports port 488 # gss-http
acl Safe_ports port 591 # filemaker
acl Safe_ports port 777 # multiling http

acl step1 at_step SslBump1
#
# Recommended minimum Access Permission configuration:
#
http_port 3128
```

```

acl Safe_ports port 777          # multiling http
acl step1 at_step SslBump1
#
# Recommended minimum Access Permission configuration:
#
http_port 3128
http_port 3129 ssl-bump cert=/etc/pki/tls/certs/ca.crt key=/etc/pki/tls/private/ca.key generate-host-certificates=on dynamic_cert_mem_cache_size=4MB
sslcrtd_program /usr/lib64/squid/security_file_certgen -s /var/lib/ssl_db -M 4MB

ssl_bump peek step1
ssl_bump bump all
# Deny requests to certain unsafe ports
http_access deny !Safe_ports

# Deny CONNECT to other than secure SSL ports
http_access deny CONNECT !SSL_ports

# Only allow cachemgr access from localhost
http_access allow manager localhost
#http_access allow localhost manager
http_access deny manager

```

```

# We strongly recommend the following be uncommented to protect innocent
# web applications running on the proxy server who think the only
# one who can access services on "localhost" is a local user
#http_access deny to_localhost

#
# INSERT YOUR OWN RULE(S) HERE TO ALLOW ACCESS FROM YOUR CLIENTS
#

# Example rule allowing access from your local networks.
# Adapt localnet in the ACL section to list your (internal) IP networks
# from where browsing should be allowed
http_access allow localnet
http_access allow localhost

# And finally deny all other access to this proxy
http_access deny all

# Squid normally listens to port 3128
#http_port 3128

# Uncomment and adjust the following to add a disk cache directory.

```

```

# from where browsing should be allowed
http_access allow localnet
http_access allow localhost

# And finally deny all other access to this proxy
http_access deny all

# Squid normally listens to port 3128
#http_port 3128

# Uncomment and adjust the following to add a disk cache directory.
cache_dir ufs /var/spool/squid 100 16 256
# Leave coredumps in the first cache dir
coredump_dir /var/spool/squid

cache_replacement_policy heap LFUDA

#
# Add any of your own refresh_pattern entries above these.
#
refresh_pattern ^ftp:          1440      20%      10080
refresh_pattern -i (/cgi-bin/|\?) 0       0%        0
refresh_pattern .              0         20%      4320

```

Why I choose LFUDA ,

LFUDA keeps larger, frequently accessed objects in cache, improving the efficiency of bandwidth usage which is ideal when cache space is tight as in my case 340KB-360KB.

LRU may evict large sites too early.
It allows newer, more popular sites to replace old.

14) Configuring fedora client to use SSL Bump port

```
root@fedora:~# export http_proxy=http://proxy.csa.lk:3128
root@fedora:~# export https_proxy=http://proxy.csa.lk:3129
root@fedora:~#
```

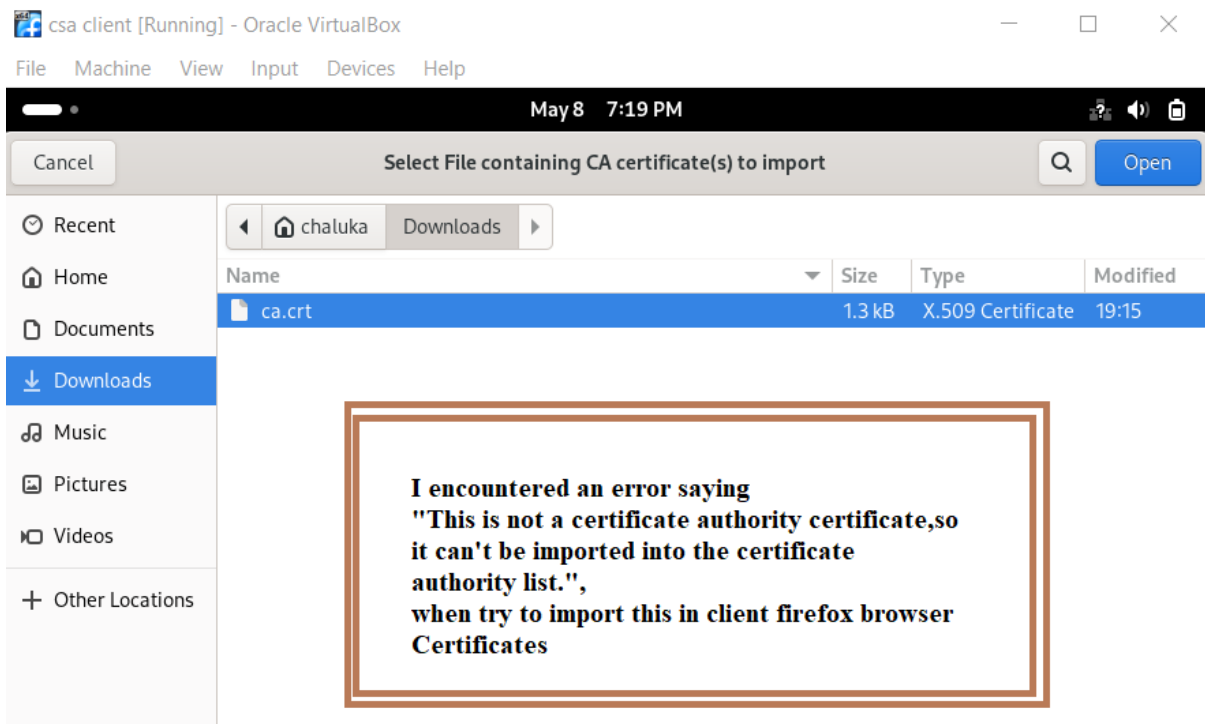
15) Trusting proxy's CA certificate on fedora – copying the CA certificate to fedora

```
chaluka@fedora:~$ scp root@192.168.1.5:/etc/pki/tls/certs/ca.crt ~/Downloads/
The authenticity of host '192.168.1.5 (192.168.1.5)' can't be established.
ED25519 key fingerprint is SHA256:E2NIgmVISx1A1/WQyd4TG1oC6TEhgfGv1MAKvtbfp90.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.5' (ED25519) to the list of known hosts.
root@192.168.1.5's password:
ca.crt                                100% 1298   403.5KB/s   00:00
chaluka@fedora:~$
```

16) Updating fedora's trusted certificate store

```
chaluka@fedora:~$ sudo cp ~/Downloads/ca.crt /etc/pki/ca-trust/source/anchors/
[sudo] password for chaluka:
chaluka@fedora:~$ sudo update-ca-trust
chaluka@fedora:~$
```

17) Error – I used the certificate that I have used previously for Apache web-server Lab.



18) Solution – I generated a proper self-signed CA certificate. I had to generate a certificate that explicitly includes CA capabilities.

*Navigating to the working directory, Generating a private key for my CA and creating a self-signed CA certificate. Common name “CSA LAB Proxy CA”

```
root@mlb-dc1-centos9:/etc/squid/ssl_cert

[root@mlb-dc1-centos9 ~]# mkdir -p /etc/squid/ssl_cert
[root@mlb-dc1-centos9 ~]# cd /etc/squid/ssl_cert
[root@mlb-dc1-centos9 ssl_cert]# openssl genrsa -out myCA.key 2048
[root@mlb-dc1-centos9 ssl_cert]# openssl req -x509 -new -nodes -key myCA.key -sha256 -days 1024 -out myCA.crt
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:
State or Province Name (full name) []:
Locality Name (eg, city) [Default City]:
Organization Name (eg, company) [Default Company Ltd]:
Organizational Unit Name (eg, section) []:
Common Name (eg, your name or your server's hostname) []:CSA LAB Proxy CA
Email Address []:
[root@mlb-dc1-centos9 ssl_cert]#
```

*Copying the files for squid

```
[root@mlb-dc1-centos9 ssl_cert]# cp myCA.crt /etc/pki/tls/certs/
[root@mlb-dc1-centos9 ssl_cert]# cp myCA.key /etc/pki/tls/private/
[root@mlb-dc1-centos9 ssl_cert]#
```

Updated the squid.conf file line accordingly,

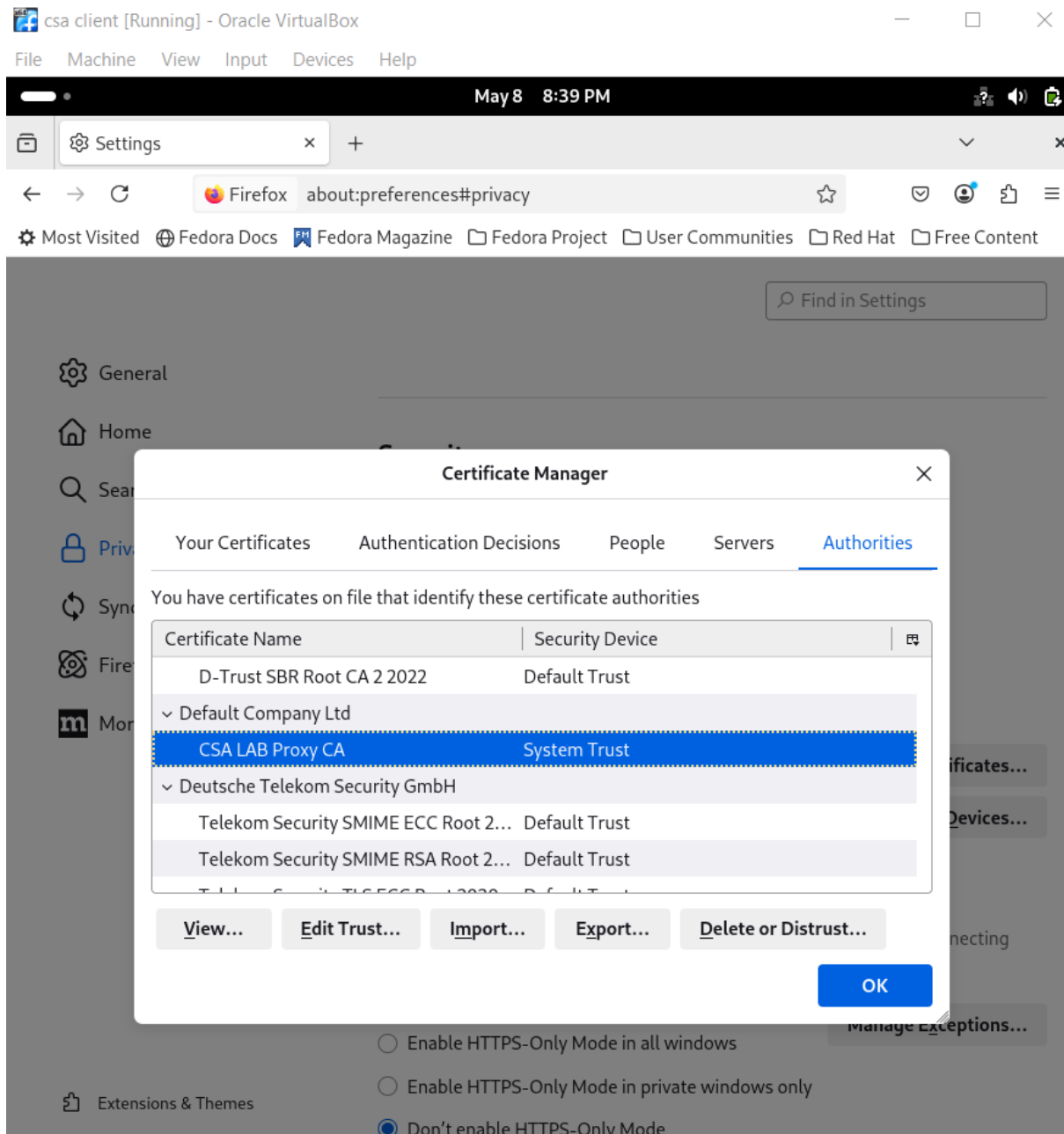
“http_port 3129 ssl-bump cert=/etc/pki/tls/certs/myCA.crt key=/etc/pki/tls/private/myCA.key
generate-host-certificates=on dynamic_cert_mem_cache_size=4MB”

19) Copying myCA.crt from server to client and trusting the CA certificate

```
chaluka@fedora:~$ scp root@192.168.1.5:/etc/pki/tls/certs/myCA.crt ~/Downloads/
root@192.168.1.5's password:
myCA.crt                                100% 1310   334.5KB/s   00:00
chaluka@fedora:~$ sudo cp ~/Downloads/myCA.crt /etc/pki/ca-trust/source/anchors/
[sudo] password for chaluka:
chaluka@fedora:~$ sudo cp ~/Downloads/myCA.crt /etc/pki/ca-trust/source/anchors/
chaluka@fedora:~$ su -
Password:
```

Then I gave this commad “sudo update-ca-trust”

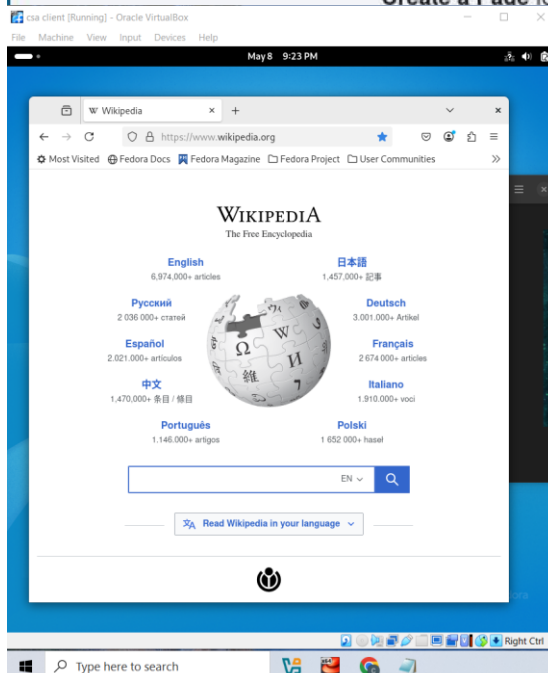
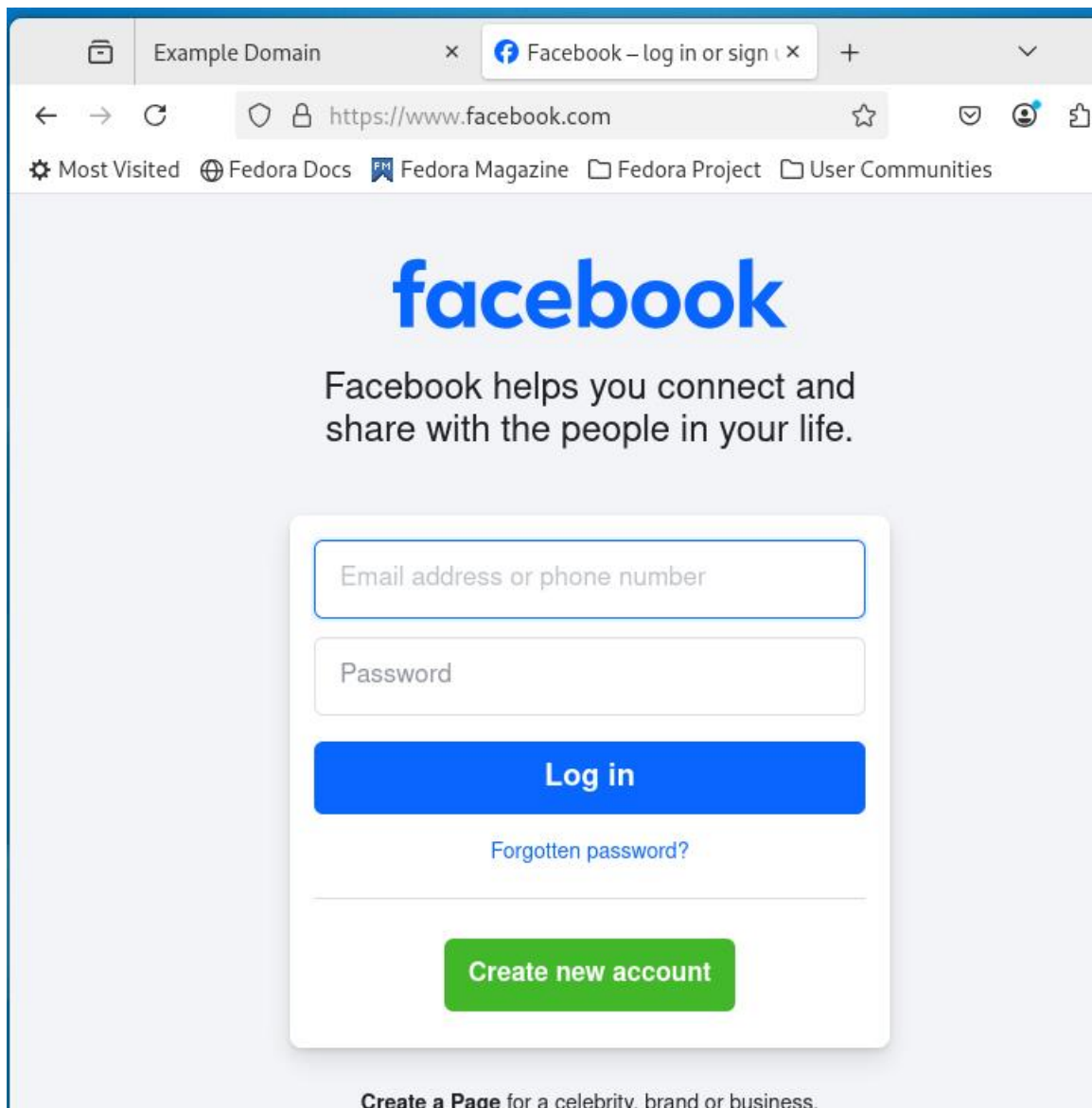
20) New certificate successfully imported.



21) Setting firewall permissions in server for https port

```
[root@mlb-dc1-centos9 ~]# firewall-cmd --permanent --add-port=3129/tcp
success
[root@mlb-dc1-centos9 ~]# firewall-cmd --reload
success
[root@mlb-dc1-centos9 ~]#
```

22) Loading facebook.com as https in browser got succeeded:



```
1746719572.149 68 192.168.1.6 TCP_MISS/200 1641 GET https://www.wikipedia.org/portal/wikipedia.org/assets/js/ 02.166.224 application/javascript
1746719572.172 99 192.168.1.6 TCP_MISS/200 19361 GET https://www.wikipedia.org/portal/wikipedia.org/assets/leg 102.166.224 image/svg+xml
1746719572.984 45 192.168.1.6 TCP_MISS/200 2361 GET https://www.wikipedia.org/static/apple-touch/wikipedia.png 0
1746719572.911 53 192.168.1.6 TCP_MISS/200 2147 GET https://www.wikipedia.org/static/favicon/wikipedia.ico - H icrosoft icon
1746719580.927 0 192.168.1.6 TCP_INM_HIT/304 371 GET https://www.wikipedia.org/ - HIER_NONE/ text/html
1746719584.619 0 192.168.1.6 TCP_INM_HIT/304 371 GET https://www.wikipedia.org/ - HIER_NONE/ text/html
1746719586.261 0 192.168.1.6 TCP_INM_HIT/304 371 GET https://www.wikipedia.org/ - HIER_NONE/ text/html
1746719587.216 0 192.168.1.6 TCP_INM_HIT/304 371 GET https://www.wikipedia.org/ - HIER_NONE/ text/html
1746719594.429 471 192.168.1.6 NONE_NONE/200 0 CONNECT www.wikipedia.org:443 - HIER_DIRECT/2001:df2:e500:edia: 1746719594.430 0 192.168.1.6 TCP_INM_HIT/304 371 GET https://www.wikipedia.org/ - HIER_NONE/ text/html
1746719606.469 429 192.168.1.6 NONE_NONE/200 0 CONNECT www.wikipedia.org:443 - HIER_DIRECT/2001:df2:e500:edia: 1746719607.811 420 192.168.1.6 TCP_INM_HIT/304 371 GET https://www.wikipedia.org/ - HIER_NONE/ text/html
1746719607.811 0 192.168.1.6 TCP_200_HIT/304 371 GET https://www.wikipedia.org/ - HIER_NONE/ text/html
```

23) Error encountered when trying to curl https sites in terminal,

This "curl https://example.com" gives this output"

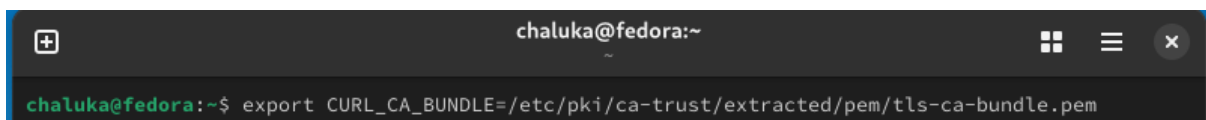
```
chaluka@fedora:~$ curl https://example.com
```

curl: (60) SSL certificate problem: self-signed certificate in certificate chain More details here: <https://curl.se/docs/sslcerts.html>

curl failed to verify the legitimacy of the server and therefore could not establish a secure connection to it. To learn more about this situation and how to fix it, please visit the webpage mentioned above.”

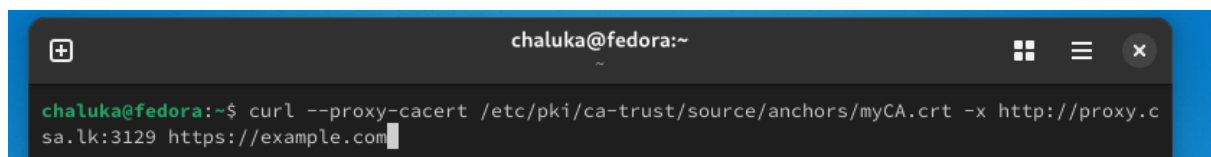
24) So I did below things,

*Set the CA Environment variable for curl



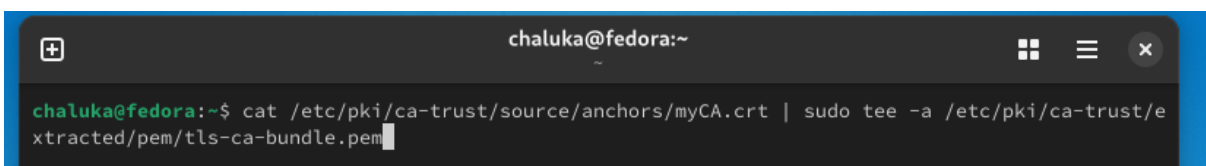
```
chaluka@fedora:~$ export CURL_CA_BUNDLE=/etc/pki/ca-trust/extracted/pem/tls-ca-bundle.pem
```

*Explicitly point to my Squid CA for Proxy

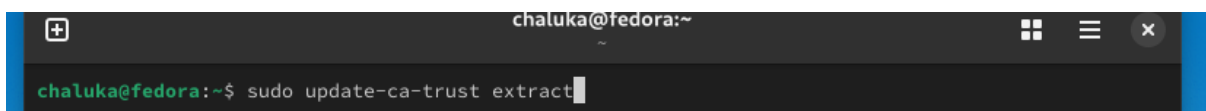


```
chaluka@fedora:~$ curl --proxy-cacert /etc/pki/ca-trust/source/anchors/myCA.crt -x http://proxy.csa.lk:3129 https://example.com
```

*Embed CA Cert into the CA Bundle



```
chaluka@fedora:~$ cat /etc/pki/ca-trust/source/anchors/myCA.crt | sudo tee -a /etc/pki/ca-trust/extracted/pem/tls-ca-bundle.pem
```



```
chaluka@fedora:~$ sudo update-ca-trust extract
```

25) Now https sites(html) can be successfully loaded in terminal through curl.

```
chaluka@fedora:~$ curl https://www.wikipedia.org
<!DOCTYPE html>
<html lang="en" class="no-js">
<head>
<meta charset="utf-8">
<title>Wikipedia</title>
<meta name="description" content="Wikipedia is a free online encyclopedia, created and edited by
volunteers around the world and hosted by the Wikimedia Foundation.">
<script>
document.documentElement.className = document.documentElement.className.replace( /(^\|s)no-js(\s|
$)/, "$1js-enabled$2" );
</script>
<meta name="viewport" content="initial-scale=1,user-scalable=yes">
<link rel="apple-touch-icon" href="/static/apple-touch/wikipedia.png">
<link rel="shortcut icon" href="/static/favicon/wikipedia.ico">
<link rel="license" href="//creativecommons.org/licenses/by-sa/4.0/">
<style>
.sprite{background-image:linear-gradient(transparent,transparent),url(portal/wikipedia.org/assets
/img/sprite-de847d1a.svg);background-repeat:no-repeat;display:inline-block;vertical-align:middle}
.svg-Commons-logo_sister{background-position:0 0;width:47px;height:47px}.svg-MediaWiki-logo_siste
r{background-position:0 -47px;width:47px;height:47px}.svg-Meta-Wiki-logo_sister{background-positi
```

```
chaluka@fedora:~$ curl https://example.com
<!doctype html>
<html>
<head>
  <title>Example Domain</title>

  <meta charset="utf-8" />
  <meta http-equiv="Content-type" content="text/html; charset=utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <style type="text/css">
    body {
      background-color: #f0f0f2;
      margin: 0;
      padding: 0;
      font-family: -apple-system, system-ui, BlinkMacSystemFont, "Segoe UI", "Open Sans", "Helv
etica Neue", Helvetica, Arial, sans-serif;
```

26) Getting website sizes to implement the caching algorithm

```
chaluka@fedora:~$ curl -I http://neverssl.com
HTTP/1.1 200 OK
Date: Fri, 09 May 2025 02:40:23 GMT
Server: Apache/2.4.62 ()
Last-Modified: Wed, 29 Jun 2022 00:23:33 GMT
ETag: "f79-5e28b29d38e93"
Accept-Ranges: bytes
Content-Length: 3961
Vary: Accept-Encoding
Content-Type: text/html; charset=UTF-8
Age: 2951
X-Cache: HIT from mlb-dc1-centos9.csa.lk
X-Cache-Lookup: HIT from mlb-dc1-centos9.csa.lk:3128
Via: 1.1 mlb-dc1-centos9.csa.lk (squid/5.5)
Connection: keep-alive

chaluka@fedora:~$ curl -I http://example.com
HTTP/1.1 200 OK
Content-Length: 1256
Content-Type: text/html
Last-Modified: Mon, 13 Jan 2025 20:11:20 GMT
ETag: "84238dfc8092e5d9c0dac8ef93371a07:1736799080.121134"
Cache-Control: max-age=1565
Date: Fri, 09 May 2025 03:29:59 GMT
Age: 0
X-Cache: HIT from mlb-dc1-centos9.csa.lk
X-Cache-Lookup: HIT from mlb-dc1-centos9.csa.lk:3128
Via: 1.1 mlb-dc1-centos9.csa.lk (squid/5.5)
Connection: keep-alive
```

```
chaluka@fedora:~$ curl -I http://httpforever.com
HTTP/1.1 200 OK
Server: nginx/1.18.0 (Ubuntu)
Date: Fri, 09 May 2025 02:41:14 GMT
Content-Type: text/html
Content-Length: 5124
Last-Modified: Wed, 22 Mar 2023 14:54:48 GMT
ETag: "641b16b8-1404"
Referrer-Policy: strict-origin-when-cross-origin
X-Content-Type-Options: nosniff
Feature-Policy: accelerometer 'none'; camera 'none'; geolocation 'none'; gyroscope 'none'; magnetometer 'none'; microphone 'none'; payment 'none'; usb 'none'
Content-Security-Policy: default-src 'self'; script-src cdnjs.cloudflare.com 'self'; style-src cdnjs.cloudflare.com 'self' fonts.googleapis.com 'unsafe-inline'; font-src fonts.googleapis.com 'self' ts.gstatic.com cdnjs.cloudflare.com; frame-ancestors 'none'; report-uri https://scotthelme.herokuapp.com/r/d/csp/enforce
Accept-Ranges: bytes
Age: 2970
X-Cache: HIT from mlb-dc1-centos9.csa.lk
X-Cache-Lookup: HIT from mlb-dc1-centos9.csa.lk:3128
```

```
</html>chaluka@fed
chaluka@fedora:~$ curl -I http://httpbin.org
HTTP/1.1 200 OK
Date: Fri, 09 May 2025 03:33:59 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 9593
Server: unicorn/19.9.0
Access-Control-Allow-Origin: *
Access-Control-Allow-Credentials: true
X-Cache: MISS from mlb-dc1-centos9.csa.lk
X-Cache-Lookup: HIT from mlb-dc1-centos9.csa.lk:3128
Via: 1.1 mlb-dc1-centos9.csa.lk (squid/5.5)
Connection: keep-alive
```

```
chaluka@fedora:~$ curl -I http://info.cern.ch
HTTP/1.1 200 OK
Date: Fri, 09 May 2025 02:46:39 GMT
Server: Apache
Last-Modified: Wed, 05 Feb 2014 16:00:31 GMT
ETag: "286-4f1aadb3105c0"
Accept-Ranges: bytes
Content-Length: 646
Content-Type: text/html
Age: 3028
X-Cache: HIT from mlb-dc1-centos9.csa.lk
X-Cache-Lookup: HIT from mlb-dc1-centos9.csa.lk:3128
Via: 1.1 mlb-dc1-centos9.csa.lk (squid/5.5)
Connection: keep-alive
```

```
chaluka@fedora:~$ curl -I http://textfiles.com/
HTTP/1.1 200 OK
Date: Fri, 09 May 2025 03:12:39 GMT
Server: Apache/2.4.58 (FreeBSD) OpenSSL/1.1.1o-freebsd
Last-Modified: Sun, 03 Nov 2024 02:54:33 GMT
ETag: "2958-625f949a948e2"
Accept-Ranges: bytes
Content-Length: 10584
Content-Type: text/html
Age: 1683
X-Cache: HIT from mlb-dc1-centos9.csa.lk
X-Cache-Lookup: HIT from mlb-dc1-centos9.csa.lk:3128
Via: 1.1 mlb-dc1-centos9.csa.lk (squid/5.5)
Connection: keep-alive
```

```
chaluka@fedora:~$ curl -I http://ftp.fau.de
HTTP/1.1 200 OK
Date: Fri, 09 May 2025 04:25:21 GMT
Server: Apache/2.4.58 (Ubuntu)
Last-Modified: Fri, 09 May 2025 04:21:04 GMT
ETag: "826a-634ac4b6708a0"
Accept-Ranges: bytes
Content-Length: 33386
Vary: Accept-Encoding
Content-Type: text/html
Age: 20
X-Cache: HIT from mlb-dc1-centos9.csa.lk
X-Cache-Lookup: HIT from mlb-dc1-centos9.csa.lk:3128
Via: 1.1 mlb-dc1-centos9.csa.lk (squid/5.5)
Connection: keep-alive
```



```
chaluka@fedora:~$ curl -I http://www.thelegacy.de
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 12610
X-WS-RateLimit-Limit: 1000
X-WS-RateLimit-Remaining: 999
Date: Fri, 09 May 2025 04:42:43 GMT
Server: Apache
Last-Modified: Thu, 24 May 2018 20:01:42 GMT
ETag: "3142-56cf91ebe5580"
Accept-Ranges: bytes
Age: 14
X-Cache: HIT from mlb-dc1-centos9.csa.lk
X-Cache-Lookup: HIT from mlb-dc1-centos9.csa.lk:3128
Via: 1.1 mlb-dc1-centos9.csa.lk (squid/5.5)
Connection: keep-alive
```

```
chaluka@fedora:~$ curl -I https://example.com
HTTP/1.1 200 Connection established

HTTP/1.1 200 OK
Accept-Ranges: bytes
Content-Type: text/html
ETag: "84238dfc8092e5d9c0dac8ef93371a07:1736799080.121134"
Last-Modified: Mon, 13 Jan 2025 20:11:20 GMT
Content-Length: 1256
Cache-Control: max-age=2748
Date: Fri, 09 May 2025 05:50:48 GMT
Alt-Svc: h3=":443"; ma=93600,h3-29=":443"; ma=93600,quic=":443"; ma=93600; v="43"
Age: 13
X-Cache: HIT from mlb-dc1-centos9.csa.lk
X-Cache-Lookup: HIT from mlb-dc1-centos9.csa.lk:3128
Via: 1.1 mlb-dc1-centos9.csa.lk (squid/5.5)
Connection: keep-alive
```

```
chaluka@fedora:~$ curl -I http://www.360doc.com/index.html
HTTP/1.1 200 OK
Server: Tengine
Content-Type: text/html; charset=utf-8
Content-Length: 256119
Vary: Accept-Encoding
Date: Sat, 10 May 2025 14:45:22 GMT
Vary: Accept-Encoding
Last-Modified: Sat, 10 May 2025 14:43:01 GMT
ETag: W/"d4f2dd4b9c1db1:0"
Expires: Sat, 10 May 2025 14:48:22 GMT
Cache-Control: max-age=180
via1: nginx com01
Accept-Ranges: bytes
Ali-Swift-Global-Savetime: 1746888322
X-Cache: MISS TCP_REFRESH_MISS dirn:2:1319129277
X-Swift-SaveTime: Sat, 10 May 2025 14:45:22 GMT
X-Swift-CacheTime: 180
Timing-Allow-Origin: *
EagleId: a3b5a49517468883222055399e
Age: 9
X-Cache: HIT from mlb-dc1-centos9.csa.lk
X-Cache-Lookup: HIT from mlb-dc1-centos9.csa.lk:3128
Via: ali com main02, ens-cache21.l2sg7[734,734,200-0,M], ens-cache25.l2sg7[735,0], cache8.sg24[740,751,200-0,M], cache1.sg24[753,0], 1.1 mlb-dc1-centos9.csa.lk (squid/5.5)
Connection: keep-alive
```

```
chaluka@fedora:~$ curl -I http://icio.us
HTTP/1.1 200 OK
Date: Sat, 10 May 2025 14:47:43 GMT
Server: Apache/2.4.18 (Ubuntu)
Last-Modified: Sat, 08 Jul 2017 17:58:19 GMT
ETag: "2c39-553d21847e20d"
Accept-Ranges: bytes
Content-Length: 11321
Vary: Accept-Encoding
Content-Type: text/html
Age: 17
X-Cache: HIT from mlb-dc1-centos9.csa.lk
X-Cache-Lookup: HIT from mlb-dc1-centos9.csa.lk:3128
Via: 1.1 mlb-dc1-centos9.csa.lk (squid/5.5)
Connection: keep-alive
```

```
chaluka@fedora:~$ curl -I http://ftp.debian.org/
HTTP/1.1 200 OK
Content-Length: 1876
Server: Apache
X-Content-Type-Options: nosniff
X-Frame-Options: sameorigin
Referrer-Policy: no-referrer
X-Xss-Protection: 1
Permissions-Policy: interest-cohort=()
Last-Modified: Fri, 01 Sep 2023 16:12:16 GMT
ETag: "754-6044e6c99e42d"
X-Clacks-Overhead: GNU Terry Pratchett
Content-Type: text/html
Accept-Ranges: bytes
Date: Sat, 10 May 2025 15:03:27 GMT
X-Served-By: cache-sin-wsss1830095-SIN
X-Cache: HIT
X-Cache-Hits: 0
X-Timer: S1746889408.903217,VS0,VE1
Vary: Accept-Encoding
Age: 109
X-Cache: HIT from mlb-dc1-centos9.csa.lk
X-Cache-Lookup: HIT from mlb-dc1-centos9.csa.lk:3128
Via: 1.1 varnish, 1.1 mlb-dc1-centos9.csa.lk (squid/5.5)
Connection: keep-alive
```

Check Page Size		
URL	Page Size (Bytes)	Page Size (KB)
http://ftp.debian.org/	1876	1.876
http://httpbin.org	9593	9.593
http://textfiles.com/	10584	10.584
http://example.com	1256	1.256
http://httpforever.com	5124	5.124
http://icio.us/	11321	11.321
http://neverssl.com	3961	3.961
http://www.thelegacy.de	12610	12.61
http://info.cern.ch	646	0.646
http://ftp.fau.de	33387	33.387

Check Page Size		
URL	Page Size (Bytes)	Page Size (KB)
http://www.360doc.com/	254766	254.766

Total Size of the first 10 web pages is 333803 bytes and the size of the 11th web site (<http://icio.us>) is 11321 bytes.

So according to the algorithm,

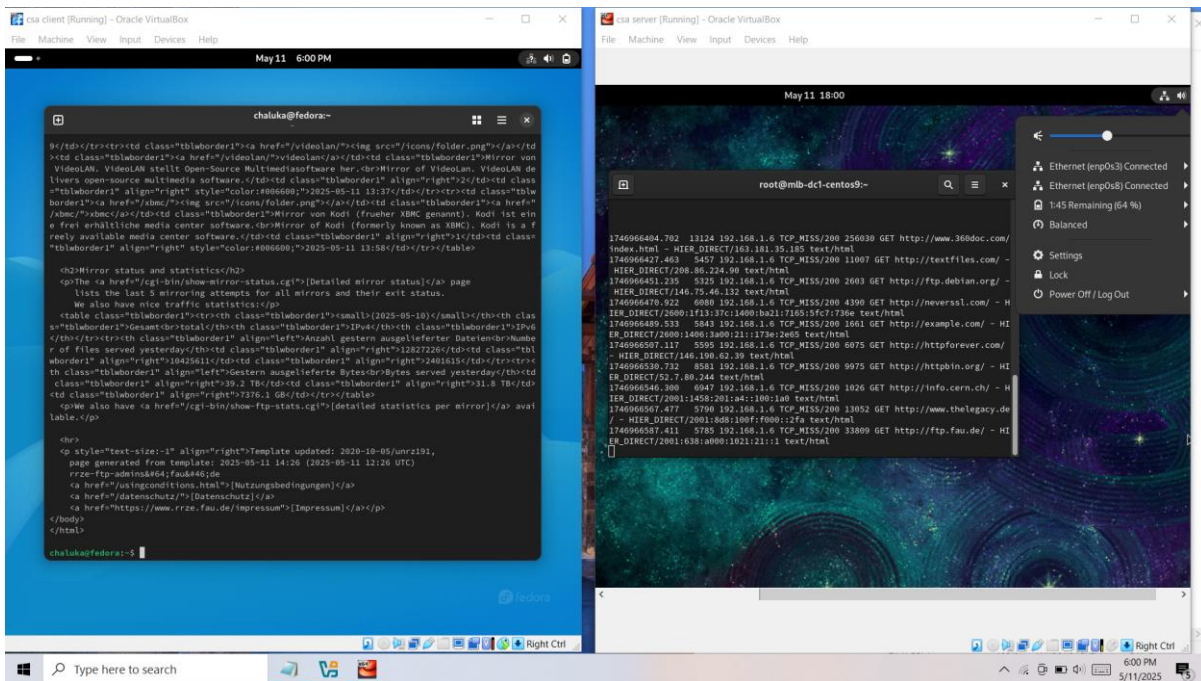
333803 bytes(total of first 10) $\leq$ CACHE folder size $<$ 345124 bytes (total of first 10 + 11th)

So I choosed 335KB as the CACHE folder size.

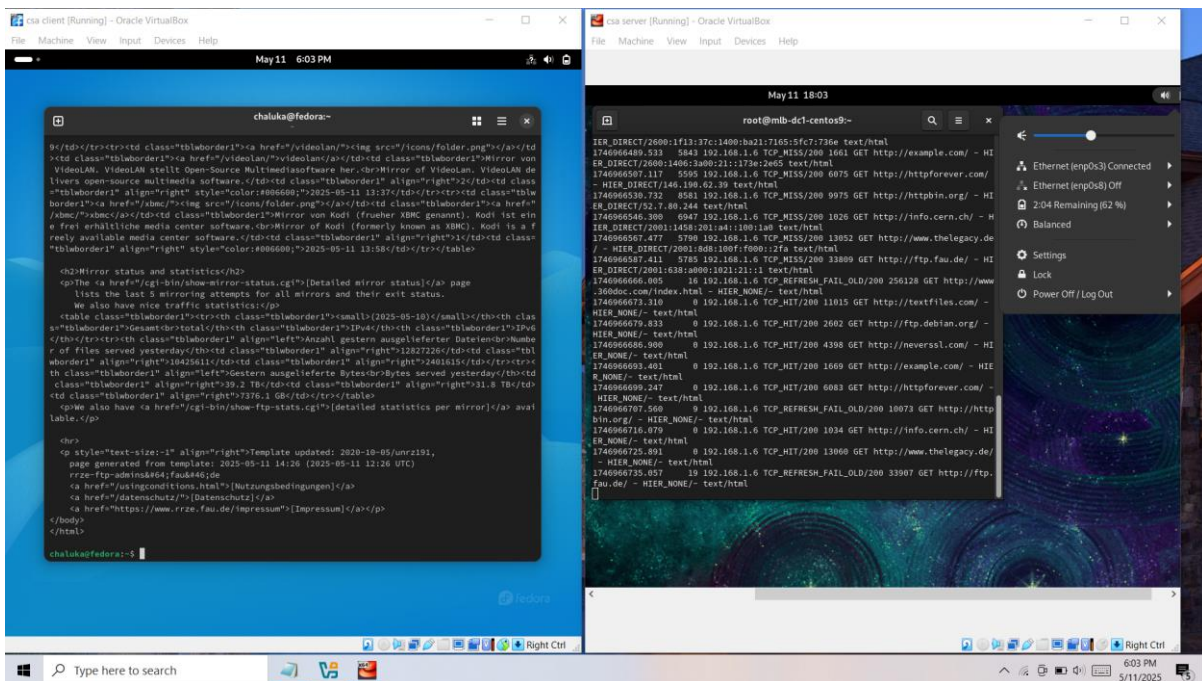
But I encountered a problem when implementing that size in squid.conf file line.

Because the minimum value I can configure is 1MB. So lets proceed with that for now.

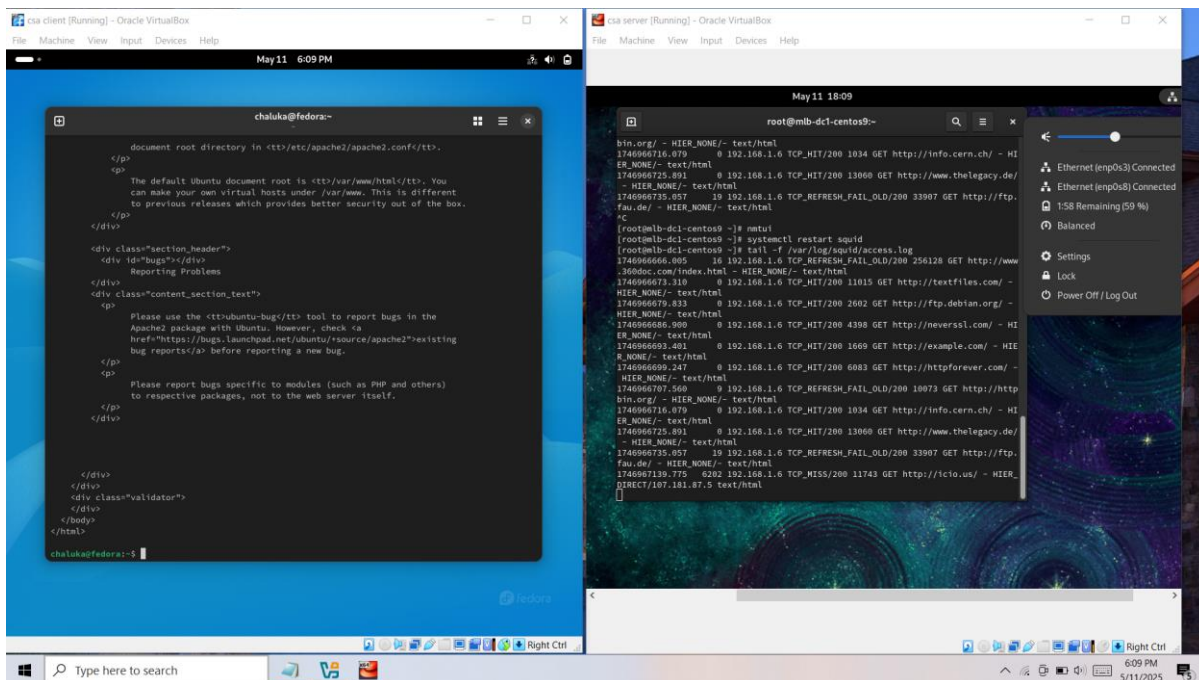
27) First 10 sites with internet



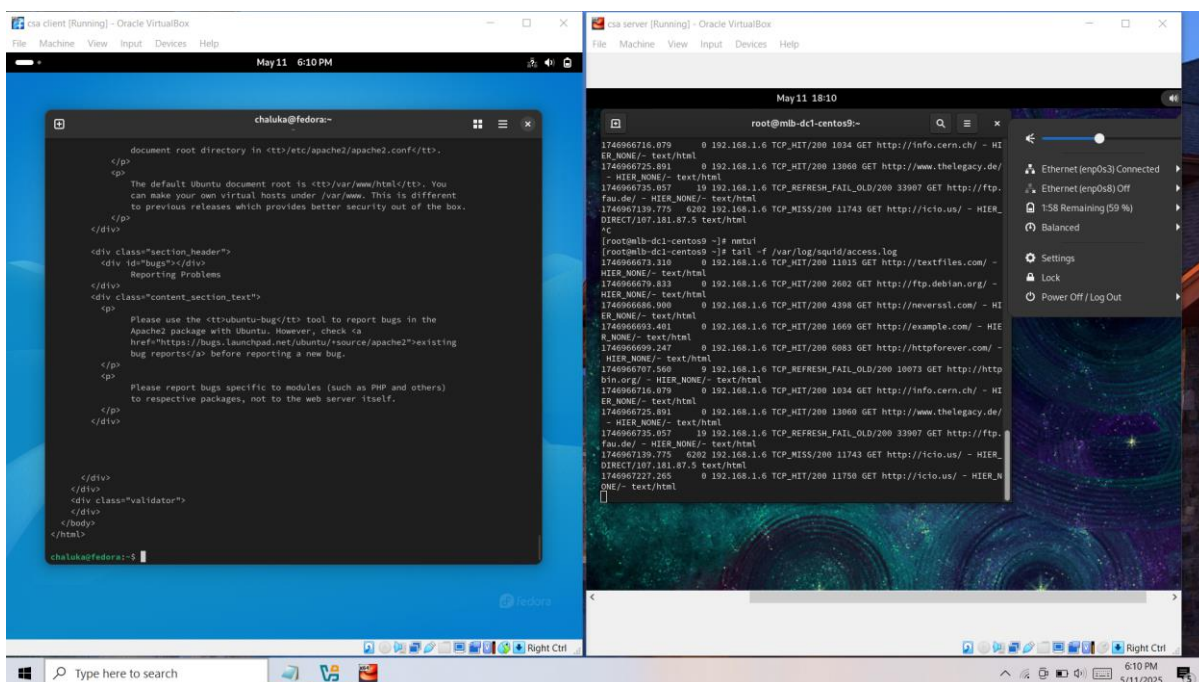
28) First 10 sites reloading without internet



29) 11th site with internet



30) 11th site without internet



To make the size exactly 335 KB I followed this approach,

31) Creating a Mount point for squid cache

```
root@mlb-dc1-centos9:~  
[root@mlb-dc1-centos9 ~]# mkdir /mnt/squidcache  
[root@mlb-dc1-centos9 ~]#
```

32) Mounting a tmpfs filesystem with 335 KB limit

```
[root@mlb-dc1-centos9 ~]# mount -t tmpfs -o size=335k tmpfs /mnt/squidcache  
[root@mlb-dc1-centos9 ~]#
```

33) Replacing cache_dir line in squid.conf file

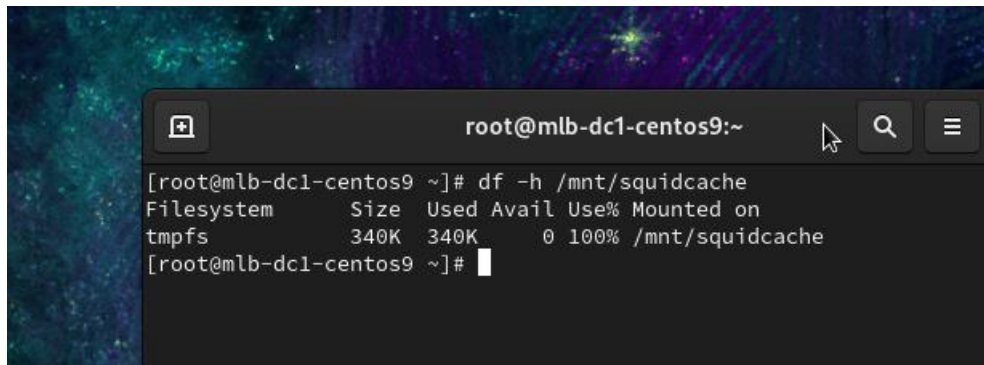
```
#  
# Uncomment and adjust the following to add a disk cache directory.  
cache_dir ufs /mnt/squidcache 1 16 256  
# Leave coredumps in the first cache dir  
coredump_dir /var/spool/squid  
  
cache_replacement_policy heap LFUDA
```

34) Rebuilding squid cache directory

```
root@mlb-dc1-centos9:~  
2025/05/11 19:18:36 kid1| Set Current Directory to /var/spool/squid  
2025/05/11 19:18:36 kid1| Creating missing swap directories  
2025/05/11 19:18:36 kid1| /mnt/squidcache exists  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/00  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/01  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/02  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/03  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/04  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/05  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/06  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/07  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/08  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/09  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/0A  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/0B  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/0C  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/0D  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/0E  
2025/05/11 19:18:36 kid1| Making directories in /mnt/squidcache/0F  
2025/05/11 19:18:36| Removing PID file (/run/squid.pid)  
  
[root@mlb-dc1-centos9 ~]# ^C  
[root@mlb-dc1-centos9 ~]# systemctl restart squid  
[root@mlb-dc1-centos9 ~]#
```

35) I loaded the first 9 sites and when I try to load the 10th site I got this error

```
"></div><div class="tl-image-b-r3c2"></div><div class="tl-image-b-r3c3"></div></div><div class="footer"><a href="/impressum.html">Impressum</a><div class="offsite-bottom"></div></div></div><script type="text/javascript" src="/the-legacy.js"></script></body></html>chaluka@fedora:~$  
chaluka@fedora:~$ curl http://ftp.fau.de  
curl: (18) end of response with 33387 bytes missing  
chaluka@fedora:~$
```



```
root@mlb-dc1-centos9:~  
[root@mlb-dc1-centos9 ~]# df -h /mnt/squidcache  
Filesystem      Size  Used Avail Use% Mounted on  
tmpfs            340K  340K    0 100% /mnt/squidcache  
[root@mlb-dc1-centos9 ~]#
```

- So according to my previous calculations 340KB was not enough. So I change it to 350K. Also I noticed here that eventhough I enabled the cache replacement policy it seems like it doesn't took any action for this. Any way I change the size to 350 KB and followed the previous steps again,

```
[root@mlb-dc1-centos9 ~]# umount /mnt/squidcache  
[root@mlb-dc1-centos9 ~]# mount -t tmpfs -o size=350k tmpfs /mnt/squid  
cache  
[root@mlb-dc1-centos9 ~]#
```

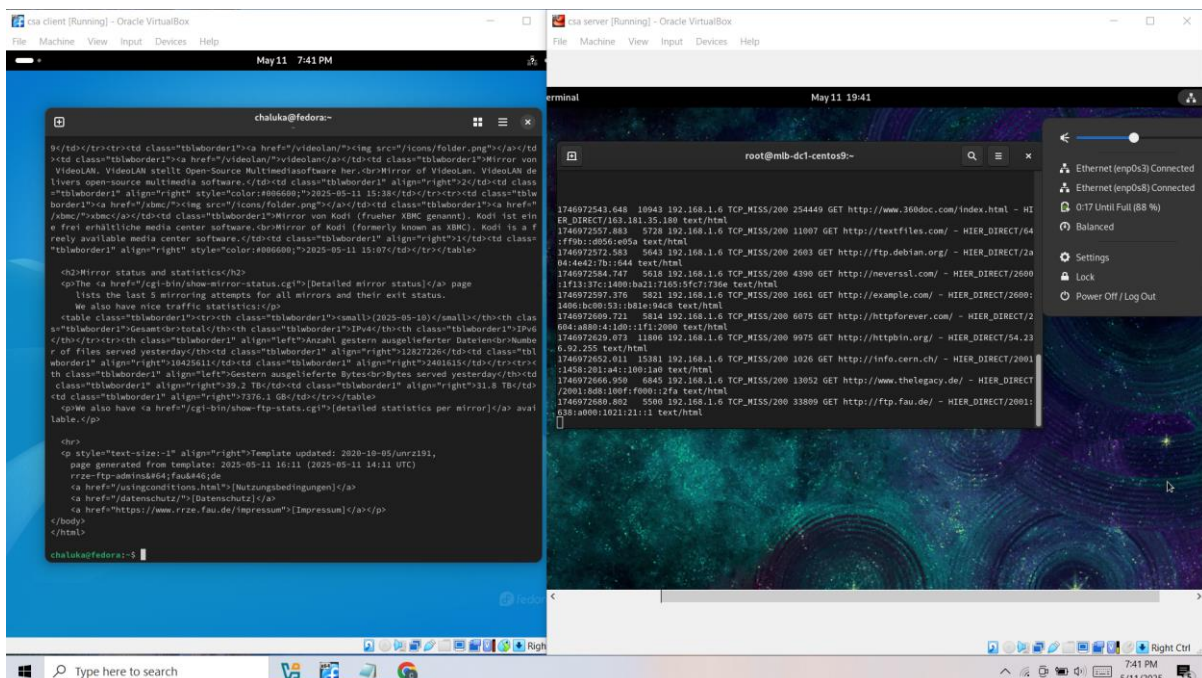
```
[root@mlb-dc1-centos9 ~]# df -h /mnt/squidcache  
Filesystem      Size  Used Avail Use% Mounted on  
tmpfs            352K    0   352K   0% /mnt/squidcache  
[root@mlb-dc1-centos9 ~]#
```

```
root@mlb-dc1-centos9:~
GNU nano 5.6.1 /etc/fstab Modified
#
# /etc/fstab
# Created by anaconda on Thu May 8 02:04:12 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/di>
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for mor>
#
# After editing this file, run 'systemctl daemon-reload' to update sy>
# units generated from this file.
#
/dev/mapper/cs_vbox-root / xfs defaults
UUID=c6d2815b-99d6-4322-90db-a2d551180e85 /boot xfs
/dev/mapper/cs_vbox-swap none swap defaults
# I added this line bla
tmpfs /mnt/squidcache tmpfs size=350k 0 0
#
```

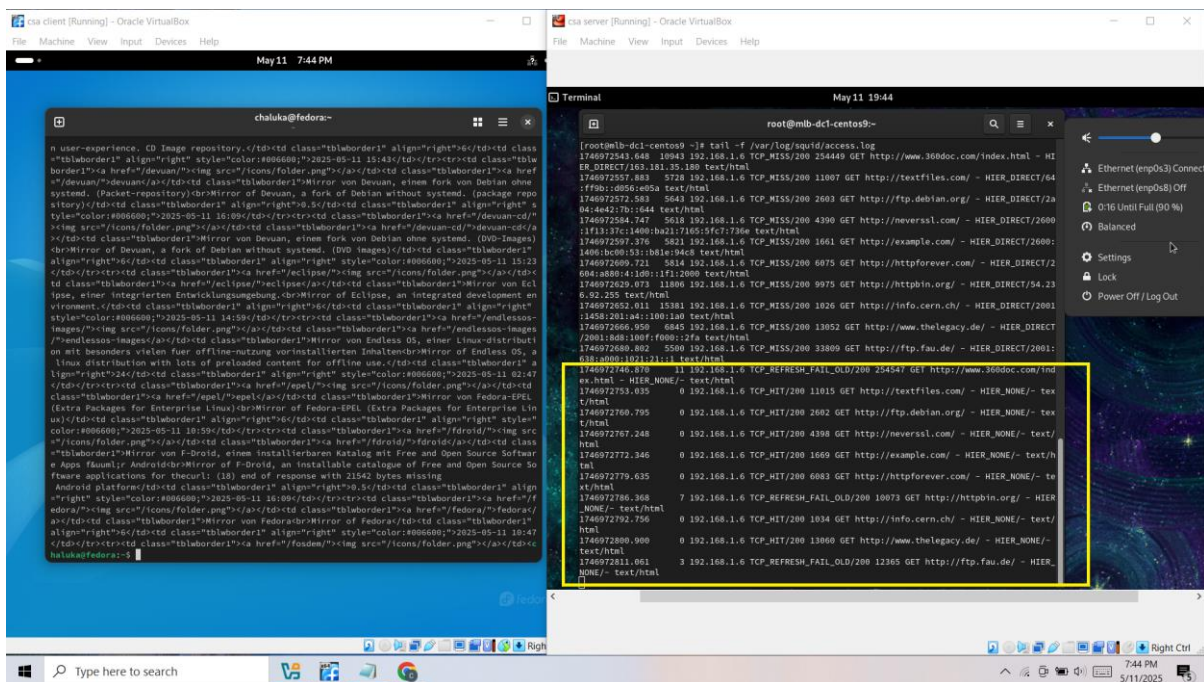
```
[root@mlb-dc1-centos9 ~]# systemctl daemon-reload
[root@mlb-dc1-centos9 ~]# mount -a
[root@mlb-dc1-centos9 ~]#
```

Now I can cache the first 10 sites successfully

36) First 10 sites with internet



37) First 10 without internet



38) Finally this is how my squid.conf file looked,

```
May 11 20:00
root@mlb-dc1-centos9:~
http_access deny manager

# We strongly recommend the following be uncommented to protect innocent
# web applications running on the proxy server who think the only
# one who can access services on "localhost" is a local user
#http_access deny to_localhost

#
# INSERT YOUR OWN RULE(S) HERE TO ALLOW ACCESS FROM YOUR CLIENTS
#

# Example rule allowing access from your local networks.
# Adapt localnet in the ACL section to list your (internal) IP networks
# from where browsing should be allowed
http_access allow localnet
http_access allow localhost

# And finally deny all other access to this proxy
http_access deny all

# Squid normally listens to port 3128
#http_port 3128
#
cache_mem 8 MB
maximum_object_size 10 MB
maximum_object_size_in_memory 0.5 KB
#
# Uncomment and adjust the following to add a disk cache directory.
cache_dir ufs /mnt/squidcache 1 16 256
# Leave coredumps in the first cache dir
coredump_dir /var/spool/squid

cache_replacement_policy heap LFUDA

#
# Add any of your own refresh_pattern entries above these.
#
refresh_pattern ^ftp:          1440      20%      10080
refresh_pattern -i (/cgi-bin/|\?) 0        0%        0
refresh_pattern .              0         20%      4320
~
```

39) Issue,

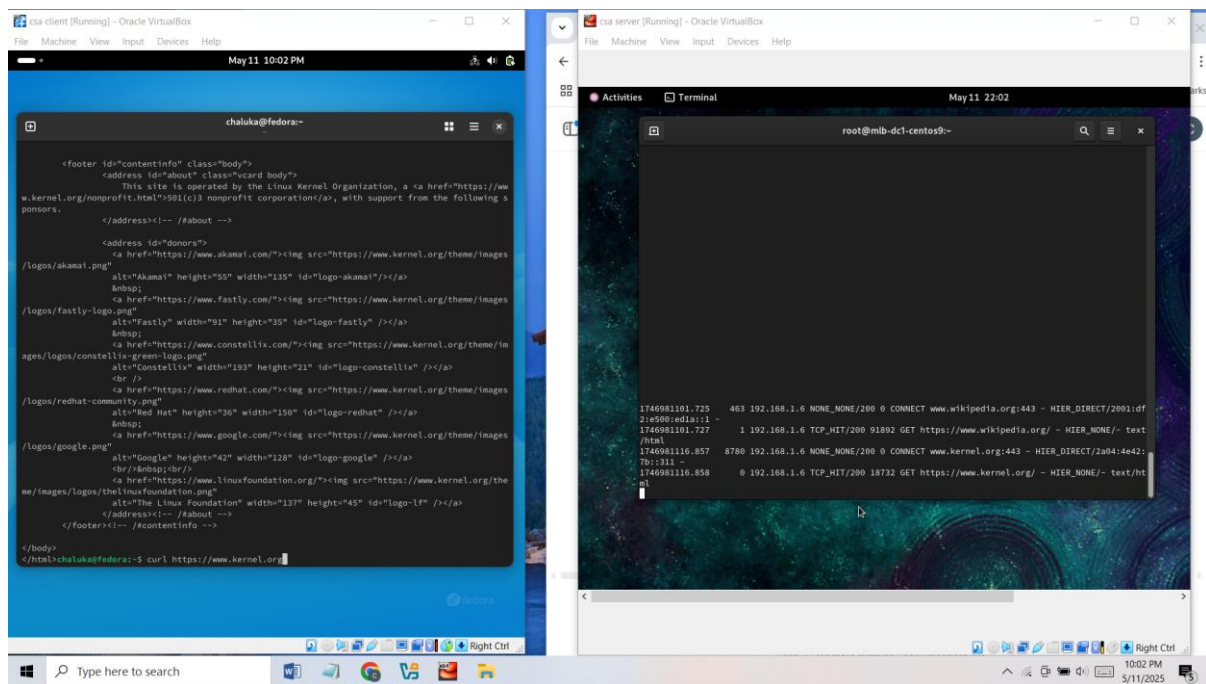
This line means “**cache_dir ufs /mnt/squidcache 1 16 256**” allocate 1MB, but the real limit is 350 KB because I mounted with,

“mount -t tmpfs -o size=340k tmpfs /mnt/squidcache”.

So I think files system fills up before squid can evict items from cache(because $350KB < 1MB$)(Squid has no idea about 350KB, it targets for 1MB).

So the solution that I can think is to start using the default directory (/var/spool/squid) to store the cached contents and specially select different 10 web pages which have total size exceeding 1MB.

40) https sites also got TCP_HIT



41) Squid.conf file content explanation

http_port 3129 ssl-bump -> Opens port 3129 for handling https connections and enables ssl Bumping

cert=/etc/pki/tls/certs/myCA.crt -> Points to my CA certificate used to generate fake site certs for clients.

key=/etc/pki/tls/private/myCA.key -> The private key corresponding to that CA certificate.

generate-host-certificates=on -> Tells Squid to dynamically generate fake site certificates on the fly for ssl bumping.

dynamic_cert_mem_cache_size=4MB -> Allocates 4 MB RAM for caching dynamically generated fake certs.

sslcrtd_program /usr/lib64/squid/security_file_certgen -s /var/lib/ssl_db -M 4MB ->

Specifies the helper tool (security_file_certgen) which generates the certificates and its database location.

acl step1 at_step SslBump1 -> Creates an acl called step1 which matches the first step of the ssl handshake.

ssl_bump peek step1 -> At step1, peek at the certificate to decide what to do (not decrypting yet).

ssl_bump bump all -> After peeking, bump (decrypt) all connections by enabling caching, filtering, and inspection of https traffic.

http_access allow all -> Allows all clients to use the proxy without restriction.(just for this assignment(not suitable for real-world)).

So from here I mark the end of my assignment report (Implement a Caching Proxy with SSL in a virtualized environment) with proven results of,

- Configuring the latest version of squid in a linux OS
- Accessing internet through squid proxy server
- Enabling caching
- Enabling appropriate cache replacement policy
- Creating a resource record for proxy in my DNS server
- Implementing SSL in the proxy environment.
- Describing troubleshooting or debugging mechanisms I got
- Suggesting solutions

I would greatly appreciate any Feedback for improvement.

Thank you for all those who supported !